

CSCI 4200: Operating System Engineering

Assignment 4 — Process: fork()

Due Date: May 3, 2026, 23:55

In this assignment, you will complete the programs provided in Assign04-Prog.zip to implement **fork()**.

In the buggy **fork()** implementation discussed in lecture, the child's kernel stack is not properly initialized: the child's stack pointer is incorrectly set by directly copying the kernel stack pointer value from the parent's task structure.

Your task is to correct this behavior by:

- **Duplicating the parent's kernel stack** into the child's kernel stack.
- **Setting the child's stack pointer correctly** to reference its own kernel stack.

Note: The syscall routine in `head.s` has been modified to facilitate your implementation and ensure the child's return from `do_fork()` as shown below.

```
# boot/head.s
....

.global current_esp_in_syscall
current_esp_in_syscall: .long 0

.global syscall_ret
syscall:
    push %ds
    pushl %edx
    pushl %ecx
    pushl %ebx
    movl $0x10,%edx
    mov %dx,%ds
    mov %dx,%es
    movl %esp, current_esp_in_syscall
    call sys_call_table(, %eax, 4)
syscall_ret:
    popl %ebx
    popl %ecx
    popl %edx
    pop %ds
    iret

.global child_fork_ret
child_fork_ret:
    movl $0, %eax    # child fork -> return 0
    jmp syscall_ret  # Jump back to pop up registers
...
```

You are specifically required to add the necessary code in the function **do_fork()** located in **process/main.c**. The kernel stacks have already been allocated in **boot/head.s**, as shown below:

```

# boot/head.s
....

# kernel stack for task0 (parent)
.align 4096
    .fill 1024,4,0
stack0_krn_ptr:
    .long 0

# kernel stack for task1 (child)
.align 4096
    .fill 1024,4,0
stack1_krn_ptr:
    .long 0

...

```

Grading Policy: The correctness of your implementation will be verified using the user-level program `mytask.c`. The expected output from `mytask.c` must match the following:

```

Bochs x86 emulator, http://bochs.sourceforge.net/
Flex86/Bochs UGABios current-cvs 07 Jan 2008
This UGA/UBE Bios is released under the GNU LGPL

Please visit :
. http://bochs.sourceforge.net
. http://www.nongnu.org/ugabios

Bochs UBE Display Adapter enabled

Bochs BIOS - build: 02/13/08
$Revision: 1.194 $ $Date: 2007/12/23 19:46:27 $
Options: apmbios pcibios eltorito rombios32

ata0 master: Generic 1234 ATA-6 Hard-Disk ( 79 MBytes)

Booting from Floppy...

Before fork: Parent is running; PID:0x0
Current pid is 0x0

After fork: ret value is 0x1
After fork: Parent-spinning; PID:0x0

After fork: ret value is 0x0
After fork: Child-spinning; PID:0x1

```

Submission Instructions

Submit a single file containing the revised source file: **`process/main.c`** with your completed implementation of **`do_fork()`**.